

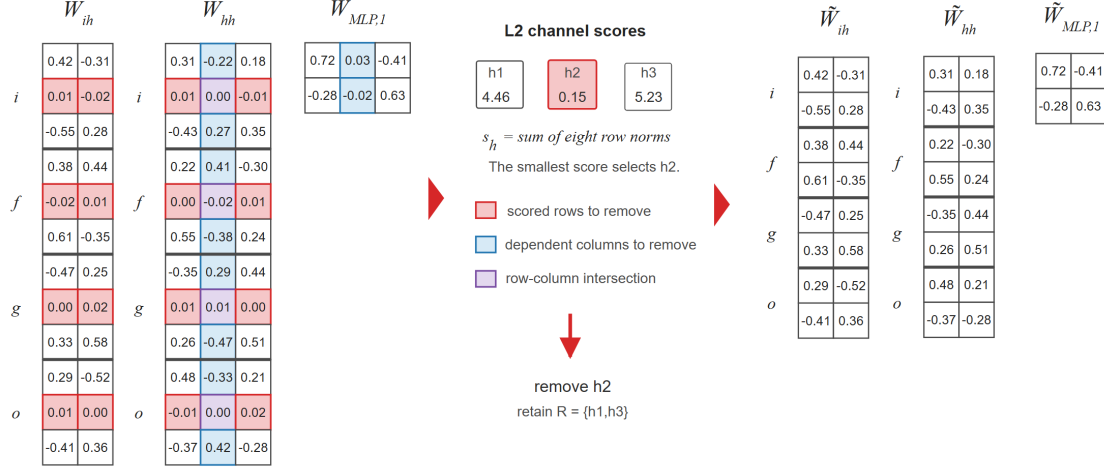


Figure 7.4: Illustrative numerical example of gate-consistent LSTM pruning. Red cells identify the rows included in the L2 saliency score, blue cells identify dependent columns that must be removed with the selected channel, and purple cells belong to both groups. Removing the lowest-scoring channel h_2 and retaining $\mathcal{R} = \{h_1, h_3\}$ produces the smaller dense tensors shown on the right. The numerical values are provided to make the scoring principle transparent, while the evaluated SOC and SOH estimators apply the same operations to their full hidden dimensions.

SOC is reduced from 64 to 45 units and SOH from 128 to 90 units, in both cases approximately 30 %. This fraction is one moderate operating point rather than the result of an exhaustive rate search. It was chosen to provide a substantial dimensional reduction while retaining useful accuracy. Determining an optimum would require separate pruning and fine-tuning runs for several candidate fractions.

The transformation is a one-shot structured removal followed by brief post-pruning fine-tuning. It is not an iterative pruning schedule and does not use pruning-aware training. The L2 score was selected because it is deterministic once the model is trained, groups the four gate rows consistently, and maps directly to smaller dense arrays. Gradient-based criteria would additionally require training data and a backward pass, while activation-based scores would depend on a calibration stream and a temporal aggregation rule. No experimental ranking of these alternatives was performed, so the L2 criterion is treated as a reproducible implementation choice rather than a universally optimal method. Figures A.4 and A.10 document the selected channels and the scope of this rationale.